$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & -\frac{3\,\mathrm{Y}_1\,k^2}{2} & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+\,\alpha\beta} & \mathcal{K}^{\#2}_{1^+\,\alpha\beta} & \mathcal{K}^{\#1}_{1^-\,\alpha} & \mathcal{K}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & -k^2\,\overset{(4)}{\kappa}_3 & \frac{k^2\,\overset{(4)}{\kappa}_4}{2\,\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{k^2\,\overset{(4)}{\kappa}_4}{2\,\sqrt{2}} & -\frac{\mathrm{Y}_2\,k^2}{2} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -k^2\,\overset{(4)}{\kappa}_2 & -\sqrt{2}\,k^2\,\overset{(4)}{\kappa}_2 \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\sqrt{2}\,k^2\,\overset{(4)}{\kappa}_2 & -2\,k^2\,\overset{(4)}{\kappa}_2
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+\,\alpha\beta} & \mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{\mathrm{Det}\,(0^+)} & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+\,\alpha\beta} & \mathcal{J}^{\#2}_{1^+\,\alpha\beta} & \mathcal{J}^{\#1}_{1^-\,\alpha} & \mathcal{J}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{\mathrm{Y}_2\,k^2}{2\,\mathrm{Det}\,(1^+)} & -\frac{k^2\,\overset{(4)}{\kappa}_4}{2\,\sqrt{2}\,\mathrm{Det}\,(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{k^2\,\overset{(4)}{\kappa}_4}{2\,\sqrt{2}\,\mathrm{Det}\,(1^+)} & -\frac{k^2\,\overset{(4)}{\kappa}_3}{\mathrm{Det}\,(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{1}{9\,k^2\,\overset{(4)}{\kappa}_2} & -\frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_2} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{\sqrt{2}}{9\,k^2\,\overset{(4)}{\kappa}_2} & -\frac{2}{9\,k^2\,\overset{(4)}{\kappa}_2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+\,\alpha\beta} & \mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

Abbreviations used in matrices

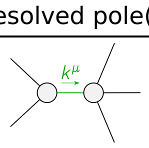
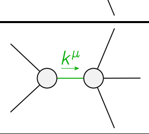
$$\mathrm{Y}_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2\ \&\&\ \mathrm{Y}_2 == 9\,\overset{(4)}{\kappa}_2 - \overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_4\ \&\&$$
$$\mathrm{Det}\,(0^+) == -\frac{3}{2}\,k^2\,(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2)\ \&\&\ \mathrm{Det}\,(1^+) == \frac{1}{8}\,k^4\,(36\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3 - (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)$$

Lagrangian

$$\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \overset{(4)}{\kappa}_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} -$$
$$6\,\overset{(4)}{\kappa}_2\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{9}{2}\,\overset{(4)}{\kappa}_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\,\overset{(4)}{\kappa}_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{0^-} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1\,\alpha}_{1^-} == \mathcal{J}^{\#2\,\alpha}_{1^-}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} ==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} +$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\ \eta^{\beta\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\ \eta^{\alpha\chi}\ \partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}^{\#1\,\alpha\beta}_{2^+} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\ \eta^{\alpha\beta}\ \partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	14	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$-\frac{216\,\overset{(4)}{\kappa}_2{}^2 - 24\,\overset{(4)}{\kappa}_2\,(2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2}{\overset{(4)}{\kappa}_2\,(36\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3 - (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)}$
	1	0	$-\frac{48\,\overset{(4)}{\kappa}_2{}^2 - 8\,\overset{(4)}{\kappa}_2\,(2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2}{\overset{(4)}{\kappa}_2\,(36\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3 - (2\,\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)}$

Resolved unitarity condition(s):	$(\overset{(4)}{\kappa}_4 < 0\ \&\&\ ((\overset{(4)}{\kappa}_3 < 0\ \&\&\ (\overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3}\ \Vert\ \overset{(4)}{\kappa}_2 > 0))\ \Vert\ (\overset{(4)}{\kappa}_3 == 0\ \&\&\ \overset{(4)}{\kappa}_2 > 0)\ \Vert$ $(0 < \overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2}\ \&\&\ 0 < \overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3})\ \Vert\ (\overset{(4)}{\kappa}_3 > -\frac{\overset{(4)}{\kappa}_4}{2}\ \&\&\ 0 < \overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3})))\ \Vert$ $(\overset{(4)}{\kappa}_4 == 0\ \&\&\ ((\overset{(4)}{\kappa}_3 < 0\ \&\&\ (\overset{(4)}{\kappa}_2 < \frac{\overset{(4)}{\kappa}_3}{9}\ \Vert\ \overset{(4)}{\kappa}_2 > 0))\ \Vert\ (\overset{(4)}{\kappa}_3 > 0\ \&\&\ 0 < \overset{(4)}{\kappa}_2 < \frac{\overset{(4)}{\kappa}_3}{9})))\ \Vert$ $(\overset{(4)}{\kappa}_4 > 0\ \&\&\ ((\overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2}\ \&\&\ (\overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3}\ \Vert\ \overset{(4)}{\kappa}_2 > 0))\ \Vert$ $(\overset{(4)}{\kappa}_3 == -\frac{\overset{(4)}{\kappa}_4}{2}\ \&\&\ (\overset{(4)}{\kappa}_2 < 0\ \Vert\ \overset{(4)}{\kappa}_2 > 0))\ \Vert\ (-\frac{\overset{(4)}{\kappa}_4}{2} < \overset{(4)}{\kappa}_3 < 0\ \&\&\ (\overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3}\ \Vert\ \overset{(4)}{\kappa}_2 > 0))\ \Vert$ $(\overset{(4)}{\kappa}_3 == 0\ \&\&\ \overset{(4)}{\kappa}_2 > 0)\ \Vert\ (\overset{(4)}{\kappa}_3 > 0\ \&\&\ 0 < \overset{(4)}{\kappa}_2 < \frac{4\,\overset{(4)}{\kappa}_3{}^2 + 4\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4{}^2}{36\,\overset{(4)}{\kappa}_3})))$